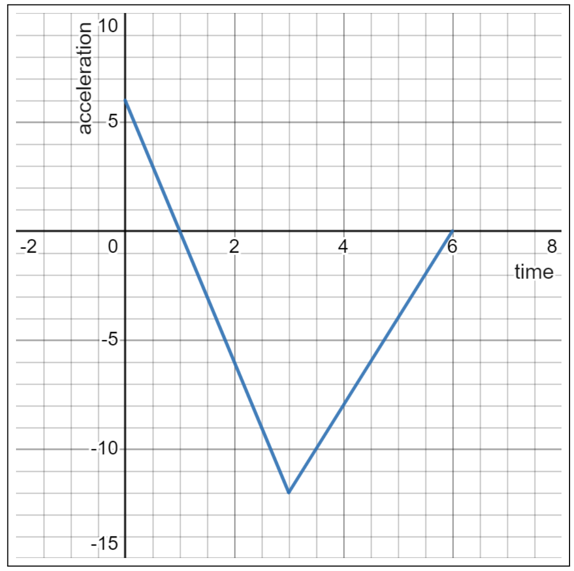
1. (Spring 2023, #9) You're driving a car at a speed of 30 ft/sec, and notice a cone ahead in the road. The graph to the right gives your acceleration t seconds after you see the cone (measured in ft/sec^2).



(a) What is your velocity at $t = 3$ seconds?

For $0 \le t \le 3$, $a(t) = 6 - 6t$

So, $v(t) = 6t - 3t^2 + C$ for some constant C.

We are given that $v(0) = 30$ (ft/sec)

$\Rightarrow C = 30$

$v(t) = 6t - 3t^2 + 30, \quad 0 \le t \le 3$

$\boxed{v(3) = 6 \cdot 3 - 3 \cdot 3^2 + 30 = 21 \text{ (ft/sec)}}$

(b) Find an expression for the velocity, $v(t)$, from $t = 3$ to $t = 6$.

From the graph of $a(t)$, $a(t) = 4t - 24$ for $3 \le t \le 6$

So $v(t) = 2t^2 - 24t + D$ for some constant D, on $3 \le t \le 6$.

Recall: $v(3) = 21$ (part (a)).

On $[3, 6]$: $v(3) = 2 \cdot 3^2 - 24 \cdot 3 + D = -54 + D$

these two numbers must evaluate to the same value

$21 = -54 + D \Rightarrow D = 75$

So: $\boxed{v(t) = 2t^2 - 24t + 75, \quad 3 \le t \le 6}$

(c) How far did the car travel from $t = 0$ to $t = 6$?

From (a) and (b), $v(t) = \begin{cases} 6t - 3t^2 + 30, & 0 \le t \le 3 \\ 2t^2 - 24t + 75, & 3 \le t \le 6 \end{cases}$ (☆)

Then $s(t) = \begin{cases} 3t^2 - t^3 + 30t + A & 0 \le t \le 3 \\ \frac{2}{3}t^3 - 12t^2 + 75t + B & 3 \le t \le 6 \end{cases}$ for some constants A, B to be determined.

Use the continuity of $s(t)$ at $t = 3$:

$[0, 3]$: $s(3) = 90 + A$

$[3, 6]$: $s(3) = 18 - 12 \cdot 9 + 75 \cdot 3 + B$

equate them $\Rightarrow B = A - 45$

So: $s(t) = \begin{cases} 3t^2 - t^3 + 30t + A & 0 \le t \le 3 \\ \frac{2}{3}t^3 - 12t^2 + 75t + A - 45 & 3 \le t \le 6 \end{cases} \Rightarrow \begin{matrix} s(0) = A \\ s(6) = A + 117 \end{matrix}$

- To find net displacement (i.e. change in position), use $\boxed{s(6) - s(0) = 117 \text{ (ft)}}$
- To find the total distance traveled, we need to consider the direction of motion over $[0, 6]$. In this problem, it turns out that $v(t)$ (☆) is always positive, meaning the car is always moving in the positive direction. So in this case, the total distance traveled = change in position = $s(6) - s(0) = 117$ (ft)

2. (Fall 2021, #1) Let $a(t)$ denote acceleration of a particle, $v(t)$ denote its velocity, and $s(t)$ denote its position. Suppose that $a(t) = 2t + 10e^{5t}$.

(a) If $v(0) = 8$, find $v(t)$.

$v(t) = t^2 + 2e^{5t} + C$ for some constant C

Use $v(0) = 8$ to find C: $v(0) = 0 + 2 \cdot 1 + C = 8 \Rightarrow C = 6$

So: $\boxed{v(t) = t^2 + 2e^{5t} + 6}$

(b) If $s(1) = 3$, find $s(t)$.

$s(t) = \frac{t^3}{3} + \frac{2}{5}e^{5t} + 6t + D$ for some constant D

Use $s(1) = 3$ to find D: $s(1) = \frac{1}{3} + \frac{2}{5}e^5 + 6 + D = 3 \Rightarrow D = -\frac{2}{5}e^5 - \frac{10}{3}$

So: $\boxed{s(t) = \frac{t^3}{3} + \frac{2}{5}e^{5t} + 6t - \frac{2}{5}e^5 - \frac{10}{3}}$

3. (Fall 2016, #7) Assume that the acceleration (in meters per second squared) of a particle at time t (in seconds) is given by

$$a(t) = \sin\left(\frac{t}{2}\right).$$

The particle is at position $x = 1$ meters at time $t = 0$ seconds and it is at position $x = 8$ meters at time $t = \pi$ seconds. Find the position $x(t)$ of the particle as a function of time. (Recall that the acceleration of a particle is the derivative of its velocity.)

Given: $x(0) = 1, \ x(\pi) = 8$

$v(t) = -2\cos\left(\frac{t}{2}\right) + C$

$x(t) = -4\sin\left(\frac{t}{2}\right) + Ct + D$ for some constants C, D

Use $x(0) = 1, \ x(\pi) = 8$ to find C, D:

$$\begin{cases} x(0) = D = 1 \\ x(\pi) = -4 \cdot 1 + C\pi + D = 8 \end{cases} \Rightarrow \begin{cases} D = 1 \\ C = \frac{11}{\pi} \end{cases}$$

So: $\boxed{x(t) = -4\sin\left(\frac{t}{2}\right) + \frac{11}{\pi}t + 1}$

4. (Fall 2011, # 10) Suppose that the acceleration of a particle (in m/sec^2) at time t (in sec) is given by the function

$$a(t) = 6t - 6$$

for $t \geq 0$.

(a) Find the velocity of $v(t)$ of the particle as a function of time if $v(0) = -9$ m/sec.

$v(t) = 3t^2 - 6t + C$ for some constant C

Use $v(0) = -9$ to find C: $v(0) = C = -9$

So $\boxed{v(t) = 3t^2 - 6t - 9}$

(b) Find the net displacement of the particle from $t = 0$ to $t = 4$, i.e., the difference in the particle's position between $t = 0$ and $t = 4$. Include units in your answer.

$s(t) = t^3 - 3t^2 - 9t + D$ for some constant D

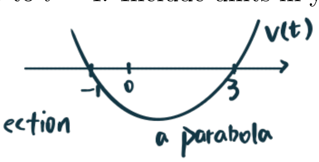
$s(0) = D$

$s(4) = 4^3 - 3 \cdot 4^2 - 9 \cdot 4 + D = -20 + D$

$\boxed{\text{net displacement} = s(4) - s(0) = -20 \text{ (m)}}$

(c) Find the total distance traveled by the particle from $t = 0$ to $t = 4$. Include units in your answer.

From (a), $v(t) = 3t^2 - 6t - 9 = 3(t-3)(t+1)$



$v(t) < 0$ on $[0, 3]$. the particle is moving in the negative direction

$v(t) > 0$ on $[3, 4]$. the particle is moving in the positive direction

To find the total distance traveled, we split $[0, 4]$ into two parts: $[0, 3]$, $[3, 4]$ and compute the displacement on each subinterval:

$s(3) = 3^3 - 3 \cdot 3^2 - 27 + D = -27 + D$

$[0, 3]$: $s(3) - s(0) = -27$

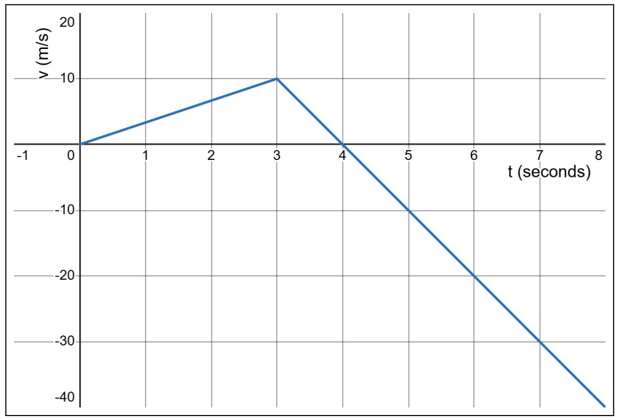
$[3, 4]$: $s(4) - s(3) = 7$

Then: total distance traveled = sum of the absolute values of the displacements

$= |s(3) - s(0)| + |s(4) - s(3)|$

$= 27 + 7 = \boxed{34 \text{ (m)}}$

5. (Fall 2002, #8) Shown below is the graph of the velocity of a toy rocket that is launched into the air, uses up all its fuel, and falls back to the ground.



(a) When does the rocket reach its highest point in the air?

$v(t) > 0$ on $(0,4)$, $v(t) < 0$ on $(0,4)$, which means the rocket's height is increasing from $t=0$ to $t=4$, then decreasing afterward.

So, $\boxed{\text{the rocket reaches its highest point at } t=4.}$

(b) How high is it at that point in time?

~~Using the point-slope formula, and Re~~

From the graph and using the point-slope formula: $v(t) = \begin{cases} \frac{10}{3}t & 0 \le t \le 3 \\ 40 - 10t & 3 \le t \le 4 \end{cases}$

So $S(t) = \begin{cases} \frac{5}{3}t^2 + C & 0 \le t \le 3 \\ 40t - 5t^2 + D & 3 \le t \le 4 \end{cases}$

Use $S(0) = 0$ to find C: $S(0) = C = 0$

Use the continuity at $t=3$: $S(3) = \frac{5}{3} \cdot 3^2 = 40 \cdot 3 - 5 \cdot 3^2 + D \Rightarrow D = -60$

$\Rightarrow S(t) = \begin{cases} \frac{5}{3}t^2 & 0 \le t \le 3 \\ 40t - 5t^2 - 60 & 3 \le t \le 4 \end{cases}$, $\boxed{\begin{aligned} S(4) &= 40 \cdot 4 - 5 \cdot 4^2 - 60 \\ &= 20 \text{ (m)} \end{aligned}}$